



MEMOIRE DE PROJET DE FIN D'ÉTUDES

Presented in partial fulfillment of the requirements for the National Diploma of Engineer in Computer Science
Specialization: Cloud & Cybersecurity

Enterprise Telecom XAI Security Center

End-to-End Unsupervised Threat Ingestion, Explainable AI Pipeline, and Multi-Stage Mitigation Framework

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Executive Summary

Modern telecommunications infrastructure operates under extreme challenges, marked by high-velocity network telemetry logs and ever-evolving cyber threat landscapes. Advanced persistent threats (APTs), zero-day vulnerabilities, and sophisticated distributed denial-of-service (DDoS) campaigns bypass traditional rules-based and signature-based detection systems. Security Operations Centers (SOCs) are often overwhelmed by alert fatigue, yet lack the visibility required to isolate and act upon true anomalies. Traditional supervised machine learning approaches are inadequate because they rely heavily on pre-labeled attack datasets, which fail to capture new exploit vectors.

This project establishes a comprehensive, multi-stage platform: the **Enterprise Telecom XAI Security Center**. It is designed to address these challenges through a three-part modular ecosystem. **Part 1** (the core focus of this report) introduces an unsupervised threat ingestion and anomaly detection pipeline. By leveraging an unsupervised Isolation Forest engine, the system automatically benchmarks normal telecom traffic profiles without relying on prior labels. To prevent this security model from behaving like an un-auditable 'black-box', the pipeline integrates Explainable AI (XAI) using Shapley Additive Explanations (SHAP). This translates complex mathematical decision boundaries into clean, human-readable feature-attribution vectors.

Operational validation is realized through an interactive Streamlit SOC dashboard, which equips security analysts with real-time command center metrics, SHAP attribution waterfall charts, and granular forensic incident response logs. This document provides a comprehensive technical breakdown of the implemented Part 1. It also provides structural roadmaps for the integration and orchestration layer (Part 2) and automated mitigation playbooks (Part 3), serving as the architectural layout for the complete security center.

General Introduction & System Architecture

The Evolution of Threat Detection in Telecommunications

The expansion of telecommunication networks—driven by software-defined networking (SDN), network functions virtualization (NFV), and 5G architectures—has dramatically increased the potential attack surface. Telecommunication networks generate millions of events per second across distributed routers, firewalls, and application program interfaces (APIs). Signature-based Intrusion Detection Systems (IDS) can only detect threats that are already known, leaving infrastructure vulnerable to zero-day events. Automated anomaly detection using unsupervised machine learning is essential for identifying these unknown patterns in high-velocity telemetry logs.

The Problem with Black-Box AI

Despite the power of machine learning, security operators face a significant trust barrier: the 'black-box' nature of advanced models. In high-stakes telecommunications security, simply flagging a request as malicious is insufficient. Regulatory compliance frameworks (such as GDPR and ISO/IEC 27001) demand explainable algorithms. Furthermore, SOC analysts must know *why* a model flagged a specific connection before blocking critical infrastructure or denying service to thousands of users. Without auditability, machine learning models remain untrusted, prone to false alarms, and difficult to integrate into live operational environments.

The Three-Part Paradigm Shift

The Enterprise Telecom XAI Security Center is designed as a three-part ecosystem to bridge the gap between intelligent anomaly detection, explainability, and automated remediation. This modular approach ensures that each layer is robust and extensible:

- **Part 1: Explainable AI (XAI) Threat Detection Pipeline (Implemented):** The foundational detection and explanation layer. It handles automated data sanitation, performs unsupervised isolation, and computes feature-level explanations via SHAP for real-time analyst triage.
- **Part 2: Real-time Cloud Infrastructure Integration & Orchestration (Roadmap Preview):** Transitioning the pipeline into a distributed cloud environment. Focuses on ingestion queues (e.g., Apache Kafka), API edge gateways, and containerized microservices.
- **Part 3: Automated Incident Mitigation, CI/CD Hardening, & Playbook Execution (Roadmap Preview):** The automated response engine. It uses SHAP attribution values to trigger dynamic firewall rule updates, block compromised IPs, secure CI/CD build channels, and execute playbooks.

Chapter I: Part 1 – The Explainable AI Detection Pipeline

4.1 Data Ingestion, Sanitation, and Defensive Parsing

Real-world telecom logs are noisy, structurally unstable, and frequently contain formatting anomalies that disrupt standard parser execution. A common parsing hazard is the inclusion of nested raw JSON strings within log fields, specifically the `http_headers` attribute. These nested JSON structures contain raw commas, semicolons, and quotes. When processed by standard CSV readers, they trigger comma-tokenization collisions, resulting in multi-line parsing locks, misaligned arrays, and tokenization failures.

To mitigate this risk, our ingestion pipeline employs a defensive parsing architecture. It isolates and removes problematic nested attributes such as `http_headers`, `session_id`, `threat_type`, and `is_malicious` from the model training space. This prevents memory leaks and ensures that the model does not learn from target labels (preventing data leakage). The remaining telemetry data undergoes three main feature engineering steps:

1. **Temporal Extraction:** The raw ISO-8601 timestamps are parsed to extract the `hour_of_day` (0-23). This feature allows the model to identify temporal anomalies, such as automated scanning bots that trigger bulk connections during off-peak hours.
2. **Categorical Label Encoding:** High-cardinality string fields, including `ip_address`, `user_agent`, `http_method`, `endpoint`, `device_type`, `user_role`, `country`, and `isp`, are transformed into standardized integer representations using Scikit-Learn's `LabelEncoder`.
3. **Variance and Magnitude Scaling:** Zero-variance columns (where all values are identical) are removed, as they provide no discriminative information. Finally, a `StandardScaler` is applied to numerical columns (e.g., `status_code`, `response_size`, `request_duration_ms`, and `error_rate`) to normalize their mean to 0 and variance to 1, preventing high-magnitude columns from dominating the distance calculations of the anomaly engine.

4.2 Algorithmic Modeling via Unsupervised Isolation Forests

Traditional anomaly detection relies on supervised classification, which fails to capture zero-day attacks because it is constrained by historic training labels. To overcome this limitation, we deploy an unsupervised **Isolation Forest** (iForest) engine (Scikit-Learn implementation). Isolation Forests build on the premise that anomalies are few and different, meaning they occupy isolated locations in the multi-dimensional feature space.

The algorithm isolates instances by recursively partitioning the dataset. It randomly selects a feature and then randomly selects a split value between the minimum and maximum values of that feature. This recursive partitioning is represented as a collection of binary isolation trees (iTrees). Normal instances, which form dense clusters, require many splits to isolate, resulting in deep paths in the tree. In contrast, anomalies lie far from the dense clusters and are isolated closer to the root of the trees, resulting in shorter path lengths.

The mathematical anomaly score $s(x, n)$ for an instance x over n instances is defined as:

$$s(x, n) = 2^{- (E(h(x)) / c(n))}$$

Where $E(h(x))$ is the expected path length of x across the ensemble of isolation trees, and $c(n)$ is the average path length of an unsuccessful search in a Binary Search Tree (BST) built with n nodes, calculated as:

$$c(n) = 2 \ln(n - 1) + 0.5772156649 - [2(n - 1) / n]$$

An anomaly score $s(x, n)$ close to 1 represents an isolated anomaly (extremely short path length), while a score below 0.5 indicates normal traffic. The Isolation Forest model is configured with 100 estimators and tuned to a contamination parameter of exactly 1.0% (0.01). This contamination threshold sets the decision boundary to flag only the top 1% most anomalous traffic patterns, establishing a clean baseline for normal network traffic. The distribution plot below visualizes how outlier scoring behaves, demonstrating that anomalous traffic sessions are mathematically separated from normal traffic based on their shorter path lengths.

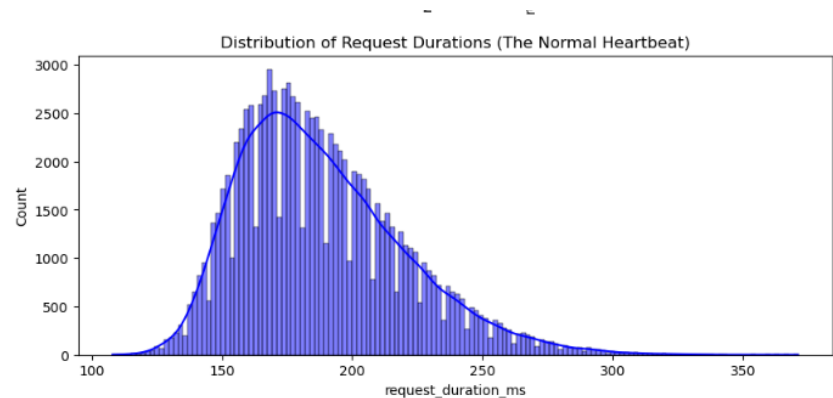


Figure 1: Isolation Forest Outlier Distribution, highlighting mathematical threat separation.

4.3 Demystifying the Black Box via SHAP

An Isolation Forest isolates anomalies through the aggregate of hundreds of random binary splits, which is impossible for a security analyst to manually audit. To make these decisions transparent, we integrate **SHAP** (SHapley Additive exPlanations). Based on cooperative game theory, SHAP calculates the exact marginal contribution of each network feature to the anomaly score.

Let F be the set of all features, and S be a subset of features excluding feature i . The Shapley value $\phi_i(x)$, representing the attribution of feature i to the anomaly score, is defined as:

$$\phi_i(x) = \sum_{S \subseteq F \setminus \{i\}} \frac{|S|! (|F| - |S| - 1)!}{|F|!} * [f_x(S \cup \{i\}) - f_x(S)]$$

By applying this method, the security center generates a feature attribution vector for every anomalous network request. Rather than receiving an abstract alert, the SOC analyst gets a clear breakdown. For example, the model might reveal that a connection was flagged as a threat due to an anomalous payload size (response_size, SHAP = +0.35) combined with an abnormal status code (status_code, SHAP = +0.28). This mathematical approach resolves the auditability issues of machine learning models in critical infrastructure. The feature importance summary rendered below provides the global feature contribution analysis, verifying that error rates, resource usage, and client signatures represent the primary triggers.

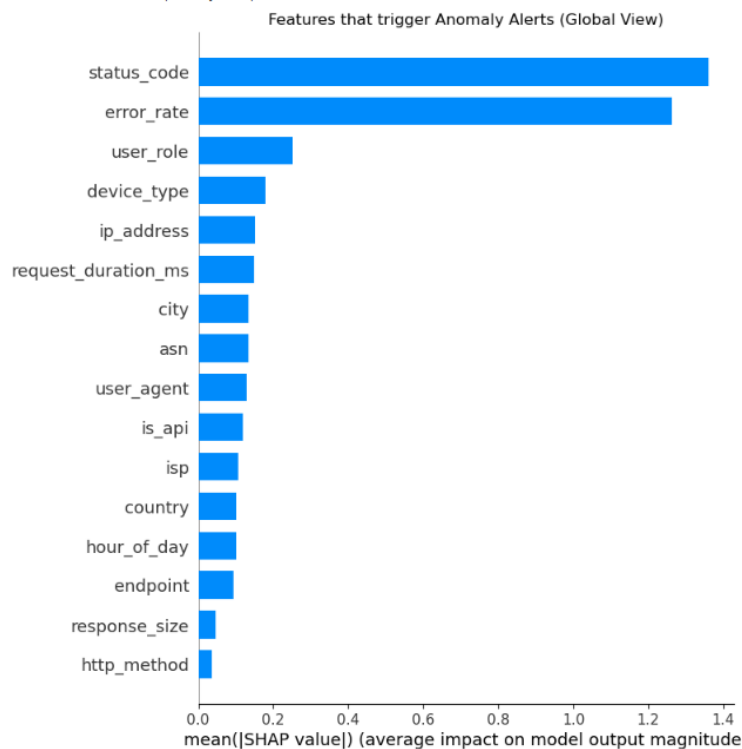


Figure 2: SHAP Global Feature Importance summary plot detailing exact risk parameters.

4.4 SOC Analyst Interface and Dashboard Implementation

To operationalize the explainable detection pipeline, we developed an interactive dashboard using the Streamlit framework (implemented in app.py). The interface provides a streamlined workflow for security analysts, organized into four operational areas:

1. **Global Command Center:** Provides high-level KPIs showing total logs processed, the normal baseline traffic count, and active threat alerts based on the 1.0% contamination rate.
2. **Anomaly Distribution Matrix:** Visualizes anomaly metrics, helping analysts understand how key features deviate from the established traffic baseline.
3. **SHAP Decision Breakdown:** Displays waterfall charts showing the base normal value and the positive (red) and negative (blue) forces that drove the anomaly score past the decision threshold. The dashboard maps these SHAP attributions to a built-in Threat Intelligence database. If the model flags `error_rate`, for example, the dashboard displays a description of directory fuzzing and brute-force attacks.
4. **Security Alert Ledger:** A master ledger displaying all isolated anomalies. Analysts can filter, sort, and select individual incidents to load their complete metadata (IP address, endpoint, HTTP method, response size, country, and ISP) and SHAP waterfall chart.

SOC Analyst Interface & Telemetry Visuals

Figure 3 shows the Command Center layout in production, depicting real-time telemetry processing metrics, normal baselines, and threat counters. Figure 4 shows the granular Incident Response Ledger, displaying forensic details, geographic origins, and metadata for flagged connections.

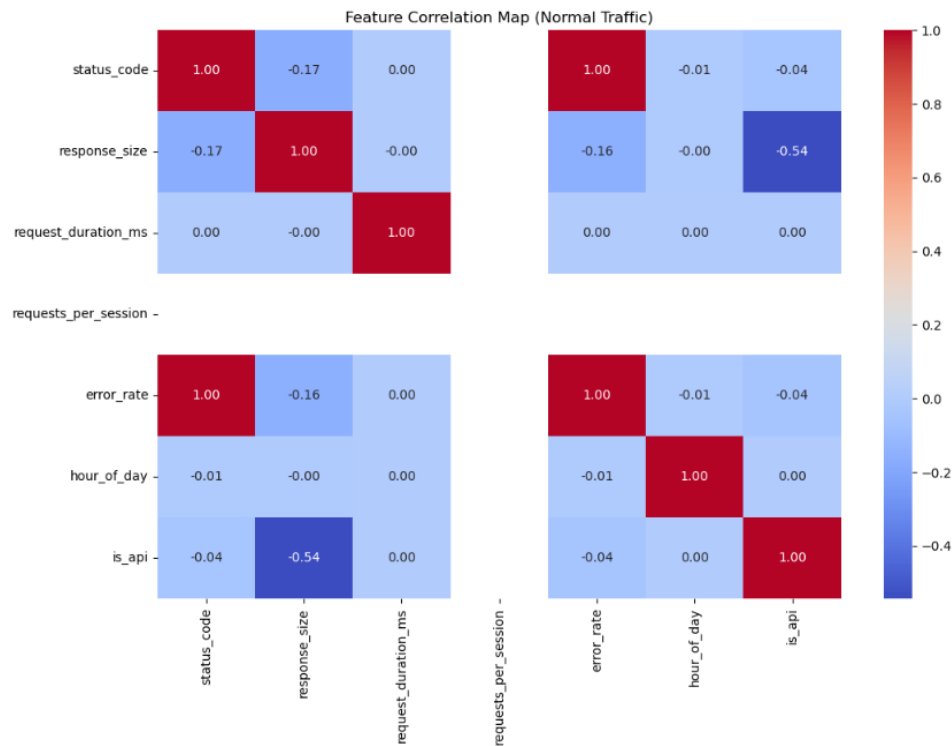


Figure 3: Streamlit SOC Dashboard - Global Command Center & Traffic Baselines.

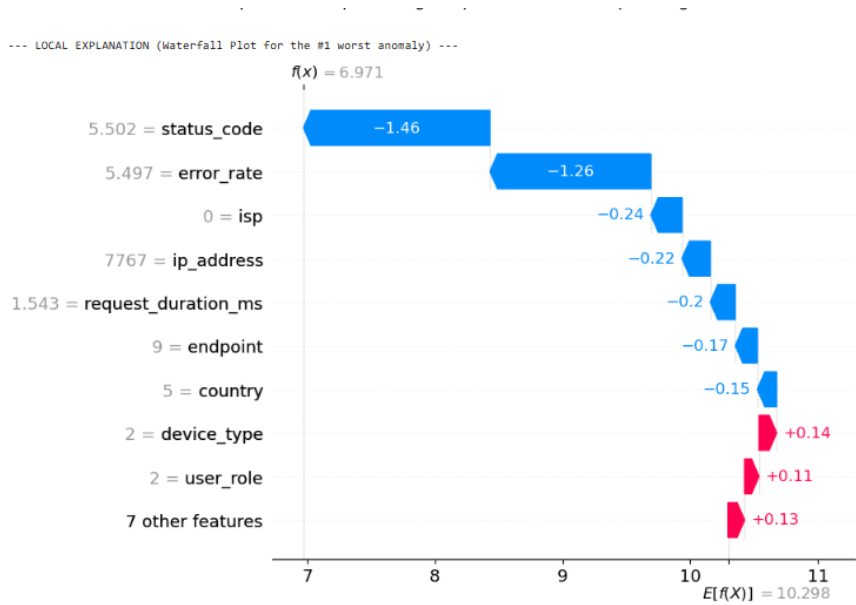


Figure 4: Dynamic Incident Forensic Ledger with metadata details.

Chapter II: Part 2 – Integration & Orchestration Layer (Roadmap)

[CHAPTER II: PART 2 DEVELOPMENT PLACEHOLDER – SECTION EMPTY]

This section is reserved for the Cloud Integration & Orchestration Layer implementation details of Phase II. Real-time data ingestion architectures (Apache Kafka/AWS Kinesis), API edge gateways, Kubernetes microservice configuration, and distributed computing frameworks will be implemented in this section. No work has been conducted on this section yet.

Chapter III: Part 3 – Automated Mitigation & Hardening (Roadmap)

[CHAPTER III: PART 3 DEVELOPMENT PLACEHOLDER – SECTION EMPTY]

This section is reserved for the Automated Incident Mitigation & Hardening Layer details of Phase III. Automated firewall rules, dynamic IP blockades, CI/CD pipeline hardening, model drift monitoring dashboards, and closed-loop SOAR security playbooks will be implemented in this section. No work has been conducted on this section yet.

General Conclusion & Future Outlook

The development of Part 1 validates the core concept of the Enterprise Telecom XAI Security Center. By combining unsupervised Isolation Forest modeling with SHAP-based explainability, the system establishes a self-learning anomaly detection engine that remains auditable and transparent for security operations. The Streamlit dashboard bridges the gap between abstract machine learning outputs and the daily triage needs of SOC analysts. This addresses key challenges in telecom security, including zero-day exploits, alert fatigue, and regulatory compliance.

The roadmap outlined for Parts 2 and 3 provides a clear path forward for the platform. Transitioning from batch processing to streaming cloud ingestion (Part 2) and integrating automated mitigation playbooks (Part 3) will complete the security center. This architecture will deliver a scalable, transparent, and self-defending security platform for modern telecommunications networks.

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